

Local and global well-posedness of the TECT gradient flow

TECT verification-first repository · claim A2-PDE-WELLPOSED

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1. Purpose and scope

This note establishes the well-posedness foundation A2 on which the entire Sector-B vacuum-selection programme rests: that the TECT free-energy minimisation has a well-defined dynamical realisation. Concretely, the L^2 -gradient flow of the canonical Brazovskii free energy is locally and globally well-posed (existence, uniqueness, parabolic regularity) for the production-anchor parameters. The argument is standard semilinear-parabolic theory (Henry 1981; Pazy 1983; Lunardi 1995) applied to the fourth-order Brazovskii operator; no TECT-specific analytic novelty is claimed, only the careful verification that the hypotheses of that theory hold here. All numerical constants are pinned by [codes/foundations/a2_wellposedness_checks.py](#) (8/8 PASS, [claims/A2-PDE-WELLPOSED/runs/a2_wellposedness_checks.json](#)).

2. Setup

Let $\mathbb{T}^3 = \mathbb{R}^3/\Lambda$ be the periodic BCC cell. The canonical free energy (A1-KERNEL-CONV convention $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$, $r_{\text{braz}} = K(q_0) = \mu^2$, with $Y > 0$) is

$$F_{\text{TECT}}[\phi] = \int_{\mathbb{T}^3} \left[\frac{1}{2} \phi K(-i\nabla)\phi + \frac{\lambda}{4} \phi^4 + \frac{\gamma}{6} \phi^6 \right] dx, \quad K(-i\nabla) = \mu^2 + Y(-\Delta - q_0^2)^2. \quad (1)$$

Its L^2 gradient flow is the semilinear parabolic Cauchy problem

$$\partial_t \phi = -\frac{\delta F_{\text{TECT}}}{\delta \phi} = -L\phi + N(\phi), \quad L := K(-i\nabla), \quad N(\phi) := -\lambda\phi^3 - \gamma\phi^5, \quad \phi(0) = \phi_0. \quad (2)$$

Work on $X := L^2(\mathbb{T}^3)$. We prove (2) is well-posed for $\phi_0 \in H^s(\mathbb{T}^3)$, $s > \frac{3}{2}$, under the standing hypotheses **(H1)** $\mu^2 > 0$ and **(H2)** $\gamma > 0$ (both hold at the production anchor $\mu^2 = 5 \times 10^{-3}$, $\gamma = 1.62$).

3. The linear part generates an analytic semigroup

L is the Fourier multiplier $K(k) = \mu^2 + Y(|k|^2 - q_0^2)^2$ on $\mathbb{T}^3$, self-adjoint with domain $D(L) = H^4(\mathbb{T}^3)$. Its spectrum is $\{K(k) : k \in \Lambda^*\} \subset [\mu^2, \infty)$, and under (H1)

$$\inf \text{spec } L = \mu^2 > 0 \quad (3)$$

(the continuum infimum μ^2 is attained on the shell $|k| = q_0$ and is a lower bound on the discrete spectrum; checked numerically, assert `spec_inf_equals_mu2_positive`). A positive self-adjoint operator is sectorial; hence $-L$ generates an analytic semigroup e^{-tL} on X , and the fractional powers L^α ($\alpha \geq 0$) are well defined with $D(L^\alpha) = H^{4\alpha}(\mathbb{T}^3) =: X^\alpha$. The analytic-semigroup smoothing estimate holds:

$$\|L^\beta e^{-tL}\|_{X \rightarrow X} \leq C_\beta t^{-\beta}, \quad C_\beta = \sup_{x \geq 0} x^\beta e^{-x} = \beta^\beta e^{-\beta} < 1, \quad (4)$$

for $0 < \beta < 1$ (assert `smoothing_exponent_finite_beta_*`: $C_{0.375} = 0.476$, $C_{0.5} = 0.429$).

4. The nonlinearity is locally Lipschitz on X^β

Fix $\beta \in (\frac{3}{8}, \frac{1}{2}]$, so $X^\beta = H^{4\beta}$ with $4\beta > \frac{3}{2}$. By the Sobolev embedding in $d = 3$, $H^{4\beta}(\mathbb{T}^3) \hookrightarrow L^\infty(\mathbb{T}^3)$ for $4\beta > \frac{3}{2}$, i.e. $\beta > \frac{3}{8}$ (assert `sobolev_linf_threshold`); moreover $H^{4\beta}$ is a Banach algebra. For ϕ, ψ in a ball $\{\|\cdot\|_{X^\beta} \leq R\}$,

$$\|N(\phi) - N(\psi)\|_X \leq (3|\lambda| \|\phi^2 + \phi\psi + \psi^2\|_{L^\infty} + 5\gamma \|\sum_{j=0}^4 \phi^j \psi^{4-j}\|_{L^\infty}) \|\phi - \psi\|_{L^2} \leq \ell(R) \|\phi - \psi\|_X, \quad (5)$$

with $\ell(R) = C(|\lambda|R^2 + \gamma R^4)$ using $\|\cdot\|_{L^\infty} \leq C_S \|\cdot\|_{X^\beta}$. Thus $N : X^\beta \rightarrow X$ is locally Lipschitz (and bounded on bounded sets). (The weaker requirement $\phi^5 \in L^2$ needs only $H^{6/5} \hookrightarrow L^{10}$, assert `sobolev_L10_threshold`; the L^∞ route at $s > \frac{3}{2}$ is taken for a clean Lipschitz constant.)

5. Local existence, uniqueness, regularity

With L sectorial (§3) and $N : X^\beta \rightarrow X$ locally Lipschitz (§4), the abstract parabolic theorem (Henry 1981, Thm. 3.3.3) applies to (2): for every $\phi_0 \in X^\beta$ there exist $\tau > 0$ and a unique mild solution

$$\phi \in C([0, \tau]; X^\beta) \cap C((0, \tau]; D(L)) \cap C^1((0, \tau]; X), \quad \phi(t) = e^{-tL} \phi_0 + \int_0^t e^{-(t-\sigma)L} N(\phi(\sigma)) d\sigma, \quad (6)$$

the integral converging by (4) ($\int_0^t (t-\sigma)^{-\beta} d\sigma < \infty$). Uniqueness and continuous dependence on ϕ_0 follow from the Lipschitz bound via Gronwall. Parabolic smoothing plus the smoothness of the polynomial N bootstraps $\phi \in C^\infty((0, \tau) \times \mathbb{T}^3)$: the solution is instantaneously smooth for $t > 0$.

6. Global existence

Along (2), F_{TECT} is a strict Lyapunov functional:

$$\frac{d}{dt} F_{\text{TECT}}[\phi(t)] = \left\langle \frac{\delta F}{\delta \phi}, \partial_t \phi \right\rangle = - \left\| \frac{\delta F}{\delta \phi} \right\|_{L^2}^2 \leq 0. \quad (7)$$

Under (H2), F_{TECT} is coercive and bounded below: by Young ($at^4 \leq bt^6 + C_*$ with $a = \frac{|\lambda|}{4}$, $b = \frac{\gamma}{12}$, $C_* = \max_t (at^4 - bt^6)$; assert `sextic_dominates_quartic_coercive`, $C_* = 1.01 \times 10^{-2}$),

$$F_{\text{TECT}}[\phi] \geq \frac{1}{2} \langle \phi, L\phi \rangle + \frac{\gamma}{12} \|\phi\|_{L^6}^6 - C_* |\mathbb{T}^3| \geq \frac{\mu^2}{2} \|\phi\|_{L^2}^2 + \frac{\gamma}{2} \|(-\Delta - q_0^2)\phi\|_{L^2}^2 + \frac{\gamma}{12} \|\phi\|_{L^6}^6 - C_* |\mathbb{T}^3|. \quad (8)$$

Hence $F[\phi(t)] \leq F[\phi_0]$ gives an a priori bound on $\phi(t)$ in $H^2(\mathbb{T}^3) \cap L^6$ (the quadratic form $\langle \phi, L\phi \rangle$ is equivalent to the H^2 norm by (H1)). Since $\beta \leq \frac{1}{2}$ implies $X^\beta = H^{4\beta} \subseteq H^2$ (assert `continuation_window_nonempty`), the X^β norm cannot blow up in finite time, so the standard continuation alternative forces $\tau = \infty$. The solution is global.

Theorem (A2). *Under (H1) $\mu^2 > 0$ and (H2) $\gamma > 0$, for every $\phi_0 \in H^s(\mathbb{T}^3)$ with $s > \frac{3}{2}$ the TECT gradient flow (2) has a unique global solution $\phi \in C([0, \infty); H^s) \cap C^\infty((0, \infty) \times \mathbb{T}^3)$, depending continuously on ϕ_0 , with $F_{\text{TECT}}[\phi(t)]$ non-increasing.*

7. Devil's-advocate (three objections)

- (α) “ q_0 incommensurate with Λ^* could push $\inf \text{spec } L$ below μ^2 .” DISMISSED: $K(k) = \mu^2 + Y(\dots)^2 \geq \mu^2$ for every k regardless of commensurability; the shell value μ^2 is a lower bound on the discrete spectrum, not a requirement that a mode sit exactly on the shell. Positivity (H1) is all that is used.

- (β) “*Global existence is convention-dependent.*” VALID with mitigation: global existence uses (H2) $\gamma > 0$ essentially — if $\gamma \leq 0$ the sextic no longer dominates the quartic and F is unbounded below, permitting finite-time blow-up. This is stated as an explicit hypothesis, not hidden; at the production anchor $\gamma = 1.62 > 0$. Local well-posedness (§5) needs only (H1).
- (γ) “*Mild solutions need not be the variational gradient flow.*” DISMISSED: the parabolic smoothing of §5 makes $\phi(t) \in D(L) = H^4$ for $t > 0$, so ϕ is a classical (strong) solution of (2) on $(0, \infty)$ and the energy identity of §6 is rigorous there; the mild formulation is used only to obtain the solution.

Quantitative sanity check (mandatory): every numerical constant is reproduced by [codes/foundations/a2_wellposedness_checks.py](#) (8/8): spectral infimum $= \mu^2 = 5 \times 10^{-3}$; Sobolev thresholds $s > 3/2$, $\beta > 3/8$; smoothing constants $C_\beta = \beta^\beta e^{-\beta} \in (0.43, 0.48)$; Young coercivity constant $C_* = 1.01 \times 10^{-2}$ at $(\lambda, \gamma) = (-0.43, 1.62)$; continuation window $\beta \in (3/8, 1/2]$ non-empty with $4\beta \leq 2$.

8. Result footer

Result ID:	A2-PDE-WELLPOSED
Precise statement:	Under $\mu^2 > 0$ and $\gamma > 0$, the TECT L2 gradient flow $d\phi/dt = -K(-i \text{ grad})\phi - \lambda \phi^3 - \gamma \phi^5$ on T^3 is globally well-posed for ϕ_0 in H^s , $s > 3/2$: unique global solution, smooth for $t > 0$, continuous dependence, F non-increasing.
Scope:	Periodic BCC cell T^3 ; production-anchor parameters; standard analytic-semigroup hypotheses (H1), (H2).
Dependencies:	A1-KERNEL-CONV (kernel convention $K = \mu^2 + Y(q^2 - q_0^2)^2$).
Hypotheses:	(H1) $\mu^2 > 0$ [sectoriality + H^2 a priori]; (H2) $\gamma > 0$ [sextic coercivity, global existence].
Evidence grade:	PROVED CONDITIONAL (analytic) + EXECUTED (8/8 checks).
Reproduction command:	python codes/foundations/a2_wellposedness_checks.py
Expected output:	8/8 PASS; exit 0; JSON artefact under claims/.../runs/.
Falsification gate:	A finite-time blow-up or non-uniqueness for some ϕ_0 in H^s ($s > 3/2$) at $\mu^2 > 0$, $\gamma > 0$; or $\inf \text{spec } L < \mu^2$.
Tier before / after:	A2 T1 / T6 PROVED CONDITIONAL (operator sign-off pending; promotion path T1→T6 as a one-shot textbook parabolic argument).
No-overclaim statement:	No claim for $\gamma \leq 0$ (blow-up possible); no claim of a unique global MINIMISER (that is vacuum selection, Sector B); no uniqueness below $s = 3/2$.
Next required action:	Operator review of the T6 promotion; then A3 (renormalisation / continuum limit) for the lattice-spacing $\rightarrow 0$ statement the flow presumes.